

University of Salento — Department of Engineering for Innovation

# Anonymous consultation of planning experts

Choosing a schedule when durations are uncertain

Booklet code: **P07**

| How to fill in (read before starting)                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fill in this file directly in Word. <b>To answer, click the boxes</b> <input type="checkbox"/> : they become <input checked="" type="checkbox"/> . Click again to undo. You will hardly need to type anything.                           |
| Set aside about <b>60 minutes without interruptions</b> . Read calmly: some questions check that the numbers in the tables have been read. Do not go back to change parts you have already completed.                                    |
| The booklet is <b>anonymous</b> : do not write your name, company or any detail that could identify you. When finished, save it and upload it to the COMPLETED folder <b>with the same file name</b> , as described in the instructions. |
| If a box does not respond to clicks (older Word versions or other programs), type an X right after the chosen box.                                                                                                                       |

☒ **I have read the information sheet, I take part voluntarily and I agree that my anonymous answers may be used in aggregate form for research purposes.**

Start time:  (the end time is entered on the last page)

*Please note your booklet code (P07) privately: you will need it only if we post a clarification question for this booklet.*

## Part 0 — About you (optional)

*The categories are deliberately broad. You may skip any question.*

### Main area of your planning work

- ☒ field service / technical assistance    ☐ distribution / transport  
☐ other    ☐ prefer not to say

### Years of experience in operational planning

- ☐ up to 5    ☐ more than 5    ☒ prefer not to say

### Do you use tools that compare several options (scenarios, multi-criteria dashboards)?

- ☐ never    ☐ sometimes    ☒ often    ☐ prefer not to say

## Part 1 — How to read the sheets (guided example)

40 work orders must be assigned to technicians for tomorrow. Regular shift: 100 time units (t.u.) per technician. Each activated technician costs 40 cost units (c.u.); each t.u. of work costs 1 c.u.; each t.u. of overtime costs an extra 1.5 c.u. Work-order durations are uncertain and vary by zone: every option was simulated on 40 plausible days plus 12 extreme days.

| Option              | Technicians | Typical-day cost | Average cost (40 days) | Average cost of the 12 worst days | "1-in-20 day" cost | Day-to-day variability | Overtime of the most loaded technician (t.u.) | Days with overtime | Cost increase under extreme conditions |
|---------------------|-------------|------------------|------------------------|-----------------------------------|--------------------|------------------------|-----------------------------------------------|--------------------|----------------------------------------|
| <b>N (standard)</b> | 5           | 728              | 849                    | 987                               | 1,058              | 91                     | 23.7                                          | 100%               | +62%                                   |
| <b>A</b>            | 6           | 750              | 818                    | 907                               | 962                | 56                     | 8.0                                           | 70%                | +54%                                   |
| <b>B</b>            | 6           | 750              | 818                    | 905                               | 951                | 54                     | 8.3                                           | 70%                | +54%                                   |
| <b>C</b>            | 7           | 790              | 837                    | 896                               | 923                | 39                     | 0.3                                           | 10%                | +38%                                   |
| <b>D</b>            | 8           | 830              | 877                    | 935                               | 963                | 38                     | 0.0                                           | 0%                 | +25%                                   |
| <b>E</b>            | 9           | 870              | 917                    | 975                               | 1,003              | 38                     | 0.0                                           | 0%                 | +24%                                   |

| Risk selector $\theta$ | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 |
|------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Recommended option     | A   | A   | A   | A   | B   | B   | C   | C   | C   |

| Column                                        | Meaning                                                                                                                                                         |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>N (standard)</b>                           | The schedule recommended by the traditional method: the cheapest on a typical day, but it ignores difficult days.                                               |
| <b>Typical-day cost</b>                       | Cost if every work order lasts exactly as planned.                                                                                                              |
| <b>Average cost (40 days)</b>                 | Average cost over the 40 simulated days.                                                                                                                        |
| <b>Average cost of the 12 worst days</b>      | Average of the 12 most expensive days out of 40 (the worst 30%).                                                                                                |
| <b>"1-in-20 day" cost</b>                     | Cost exceeded on only about one day in twenty.                                                                                                                  |
| <b>Day-to-day variability</b>                 | How far, on average, the cost of a day departs from the average cost. Low = predictable days. It does <b>not</b> tell whether technicians have equal workloads. |
| <b>Overtime of the most loaded technician</b> | Average overtime, in t.u., of the technician who works the most overtime on each day.                                                                           |
| <b>Days with overtime</b>                     | Share of days on which at least one technician exceeds the shift.                                                                                               |
| <b>Cost increase under extreme conditions</b> | How much the average cost rises on 12 days much less favourable than the simulated ones.                                                                        |
| <b>Risk selector <math>\theta</math></b>      | Low values: average cost matters most. High values: avoiding the worst days matters most. For each value the system shows the option it would recommend.        |

*Reading example: option C uses 7 technicians (one more than A and B). Compared with N it costs 62 c.u. more on a typical day but 135 c.u. less on a "1-in-20 day"; the overtime of the most loaded technician falls from 23.7 to 0.3 t.u. With the selector between 0.7 and 0.9 the system recommends C; between 0.1 and 0.4 it recommends A.*

## Part 2 — Comprehension check (on the example on the previous page)

*You may look at the example table again. Answer before turning the page.*

**Q1. Which column gives the cost of a difficult day that occurs about once every twenty days?**

- ☐ Average cost    ☒ “1-in-20 day” cost  
☐ Day-to-day variability    ☐ Typical-day cost

**Q2. Which option has the lowest typical-day cost?**

- ☒ N    ☐ A    ☐ B    ☐ C    ☐ D    ☐ E

**Q3. With the selector at 0.5, which option is recommended?**

- ☐ A    ☒ B    ☐ C    ☐ D    ☐ E

**Q4. Does low day-to-day variability mean that technicians have equal workloads?**

- ☐ Yes    ☒ No    ☐ The table does not say

**Q5. Compared with N, how does option C perform on a “1-in-20 day”?**

- ☐ it costs more    ☒ it costs less    ☐ same cost

**Solutions to Part 2 (read after answering; do not change your answers)**

- Q1: “1-in-20 day” cost — the cost exceeded on only about one day in twenty.
- Q2: N — the standard recommendation is the cheapest on a typical day (728 c.u.).
- Q3: B — in the selector row, B appears under 0.5.
- Q4: No — variability refers to days, not to how work is shared among technicians.
- Q5: it costs less — 923 c.u. versus 1,058 c.u.

*If you got an answer wrong, reread the corresponding row of Part 1 before continuing.*

**Part 3 — Your priorities**

Distribute **exactly 10 tokens** among the three criteria by clicking one box per token. More tokens = more important to you when you plan a day.

| Criterion                                                             | Tokens (click)                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Average daily cost (column “Average cost”)                            | <input checked="" type="checkbox"/> <input checked="" type="checkbox"/> <input checked="" type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/>                       |
| Containing difficult days (columns “12 worst days” and “1-in-20 day”) | <input checked="" type="checkbox"/> <input checked="" type="checkbox"/> <input checked="" type="checkbox"/> <input checked="" type="checkbox"/> <input checked="" type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> |
| Predictable days (column “Day-to-day variability”)                    | <input checked="" type="checkbox"/> <input checked="" type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/>                                  |

**Which description fits you best? (one only)**

- ☐ Budget-driven: average cost matters; the rest is secondary.
- ☐ Balanced: average cost and difficult days matter equally.
- ☒ Service level: avoiding difficult days comes before cost.
- ☐ Overtime-averse: avoid overtime and difficult days, with an eye on cost.
- ☐ Predictable days: above all I want predictable days.
- ☐ Worst case: I must protect myself against the worst days (contractual or regulatory constraints).
- ☐ None of these.

## Case 1 — Tomorrow's schedule (single recommendation)

40 work orders must be assigned to technicians for tomorrow. Regular shift: 100 time units (t.u.) per technician. Each activated technician costs 40 cost units (c.u.); each t.u. of work costs 1 c.u.; each t.u. of overtime costs an extra 1.5 c.u. Work-order durations are uncertain and vary by zone: every option was simulated on 40 plausible days plus 12 extreme days.

| Option         | Technicians | Typical-day cost | Average cost (40 days) | Average cost of the 12 worst days | "1-in-20 day" cost | Day-to-day variability | Overtime of the most loaded technician (t.u.) | Days with overtime | Cost increase under extreme conditions |
|----------------|-------------|------------------|------------------------|-----------------------------------|--------------------|------------------------|-----------------------------------------------|--------------------|----------------------------------------|
| Recommendation | 5           | 702              | 840                    | 998                               | 1,058              | 107                    | 20.4                                          | 95%                | +76%                                   |

### Reading check

**R1. How many technicians does the recommendation use?**

☐ 4   ☒ 5   ☐ 6   ☐ 7

**R2. What is the cost of a "1-in-20 day" (c.u.)?**

☐ 702   ☐ 998   ☐ 840   ☒ 1,058

**R3. By how much does the average cost increase under extreme conditions?**

☒ +76%   ☐ +20%   ☐ +95%   ☐ +51%

**D4. Which column weighed most in your decision? (one only)**

☐ Technicians   ☐ Typical-day cost  
☐ Average cost (40 days)   ☐ Average cost of the 12 worst days  
☐ "1-in-20 day" cost   ☐ Day-to-day variability  
☐ Overtime of the most loaded technician (t.u.)   ☒ Days with overtime  
☐ Cost increase under extreme conditions

### Your decision

**D1. Would you adopt this schedule for tomorrow?**

☒ Yes   ☐ Yes, but I would like to see alternatives   ☐ No

**D2. How likely is it that tomorrow's budget will be exceeded?**

very unlikely   ☐ 1   ☐ 2   ☐ 3   ☒ 4   ☐ 5   very likely

**D3. Main reasons for your answer (at most 2)**

☐ lowest typical-day cost   ☒ lowest average cost   ☐ cheaper worst days  
☐ lower day-to-day variability   ☒ fewer technicians   ☐ less overtime for the most loaded technician  
☐ better under extreme conditions   ☐ other

**D5. How confident are you in your decision?**

not at all   ☐ 1   ☐ 2   ☐ 3   ☐ 4   ☐ 5   ☐ 6   ☒ 7   completely

## Case 2 — Tomorrow's schedule (single recommendation)

40 work orders must be assigned to technicians for tomorrow. Regular shift: 100 time units (t.u.) per technician. Each activated technician costs 40 cost units (c.u.); each t.u. of work costs 1 c.u.; each t.u. of overtime costs an extra 1.5 c.u. Work-order durations are uncertain and vary by zone: every option was simulated on 40 plausible days plus 12 extreme days.

| Option         | Technicians | Typical-day cost | Average cost (40 days) | Average cost of the 12 worst days | "1-in-20 day" cost | Day-to-day variability | Overtime of the most loaded technician (t.u.) | Days with overtime | Cost increase under extreme conditions |
|----------------|-------------|------------------|------------------------|-----------------------------------|--------------------|------------------------|-----------------------------------------------|--------------------|----------------------------------------|
| Recommendation | 6           | 787              | 911                    | 1,050                             | 1,090              | 97                     | 19.9                                          | 82%                | +70%                                   |

### Reading check

**R1. How many technicians does the recommendation use?**

☐ 5   ☒ 6   ☐ 7   ☐ 8

**R2. What is the cost of a "1-in-20 day" (c.u.)?**

☐ 911   ☒ 1,090   ☐ 1,050   ☐ 787

**R3. By how much does the average cost increase under extreme conditions?**

☐ +16%   ☐ +39%   ☐ +82%   ☒ +70%

**D4. Which column weighed most in your decision? (one only)**

☐ Technicians   ☐ Typical-day cost  
☐ Average cost (40 days)   ☐ Average cost of the 12 worst days  
☒ "1-in-20 day" cost   ☐ Day-to-day variability  
☐ Overtime of the most loaded technician (t.u.)   ☐ Days with overtime  
☐ Cost increase under extreme conditions

### Your decision

**D1. Would you adopt this schedule for tomorrow?**

☒ Yes   ☐ Yes, but I would like to see alternatives   ☐ No

**D2. How likely is it that tomorrow's budget will be exceeded?**

very unlikely   ☐ 1   ☐ 2   ☐ 3   ☒ 4   ☐ 5   very likely

**D3. Main reasons for your answer (at most 2)**

☐ lowest typical-day cost   ☒ lowest average cost   ☐ cheaper worst days  
☐ lower day-to-day variability   ☐ fewer technicians   ☐ less overtime for the most loaded technician  
☐ better under extreme conditions   ☒ other

**D5. How confident are you in your decision?**

not at all   ☐ 1   ☐ 2   ☐ 3   ☐ 4   ☒ 5   ☐ 6   ☐ 7   completely

### Case 3 — Tomorrow's schedule (several options and selector)

40 work orders must be assigned to technicians for tomorrow. Regular shift: 100 time units (t.u.) per technician. Each activated technician costs 40 cost units (c.u.); each t.u. of work costs 1 c.u.; each t.u. of overtime costs an extra 1.5 c.u. Work-order durations are uncertain and vary by zone: every option was simulated on 40 plausible days plus 12 extreme days.

| Option              | Technicians | Typical-day cost | Average cost (40 days) | Average cost of the 12 worst days | "1-in-20 day" cost | Day-to-day variability | Overtime of the most loaded technician (t.u.) | Days with overtime | Cost increase under extreme conditions |
|---------------------|-------------|------------------|------------------------|-----------------------------------|--------------------|------------------------|-----------------------------------------------|--------------------|----------------------------------------|
| <b>N (standard)</b> | 6           | 796              | 951                    | 1,127                             | 1,196              | 121                    | 20.4                                          | 85%                | +79%                                   |
| <b>A</b>            | 7           | 836              | 928                    | 1,049                             | 1,117              | 79                     | 5.9                                           | 55%                | +72%                                   |
| <b>B</b>            | 8           | 876              | 951                    | 1,039                             | 1,088              | 61                     | 1.1                                           | 15%                | +56%                                   |
| <b>C</b>            | 9           | 916              | 988                    | 1,072                             | 1,105              | 59                     | 0.0                                           | 2%                 | +40%                                   |
| <b>D</b>            | 10          | 956              | 1,028                  | 1,112                             | 1,142              | 59                     | 0.0                                           | 0%                 | +37%                                   |
| <b>E</b>            | 11          | 996              | 1,068                  | 1,152                             | 1,182              | 59                     | 0.0                                           | 0%                 | +31%                                   |

| Risk selector $\theta$ | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 |
|------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Recommended option     | A   | A   | A   | A   | A   | A   | B   | B   | B   |

#### Reading check

**R1. Which option has the lowest "1-in-20 day" cost?**

☐ N ☐ A ☒ B ☐ C ☐ D ☐ E

**R2. With the selector set to 0.7, which option is recommended?**

☐ A ☒ B ☐ C ☐ D ☐ E

**R3. How many technicians does the option with the lowest average cost use?**

☐ 6 ☒ 7 ☐ 8 ☐ 9

**D4. Which column weighed most in your decision? (one only)**

☐ Technicians ☐ Typical-day cost  
☐ Average cost (40 days) ☒ Average cost of the 12 worst days  
☐ "1-in-20 day" cost ☐ Day-to-day variability  
☐ Overtime of the most loaded technician (t.u.) ☐ Days with overtime  
☐ Cost increase under extreme conditions ☐ selector row

#### Your decision

**D1. Which option would you adopt for tomorrow? (one only)**

☐ N ☐ A ☒ B ☐ C ☐ D ☐ E

**D2. If you used the selector, which value would you set?**

☐ 0.1 ☐ 0.2 ☐ 0.3 ☐ 0.4 ☐ 0.5 ☐ 0.6 ☒ 0.7 ☐ 0.8 ☐ 0.9 ☐ I would not use it

**D3. Main reasons for your choice (at most 2)**

☐ lowest typical-day cost ☐ lowest average cost ☒ cheaper worst days  
☐ lower day-to-day variability ☐ fewer technicians ☐ less overtime for the most loaded technician  
☐ better under extreme conditions ☒ I followed the selector ☐ other

**D5. How confident are you in your decision?**

not at all ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☒ 7 completely

## Case 4 — Tomorrow's schedule (several options and selector)

40 work orders must be assigned to technicians for tomorrow. Regular shift: 100 time units (t.u.) per technician. Each activated technician costs 40 cost units (c.u.); each t.u. of work costs 1 c.u.; each t.u. of overtime costs an extra 1.5 c.u. Work-order durations are uncertain and vary by zone: every option was simulated on 40 plausible days plus 12 extreme days.

| Option              | Technicians | Typical-day cost | Average cost (40 days) | Average cost of the 12 worst days | "1-in-20 day" cost | Day-to-day variability | Overtime of the most loaded technician (t.u.) | Days with overtime | Cost increase under extreme conditions |
|---------------------|-------------|------------------|------------------------|-----------------------------------|--------------------|------------------------|-----------------------------------------------|--------------------|----------------------------------------|
| <b>N (standard)</b> | 6           | 780              | 926                    | 1,076                             | 1,121              | 99                     | 22.2                                          | 88%                | +68%                                   |
| <b>A</b>            | 6           | 780              | 880                    | 1,017                             | 1,059              | 84                     | 11.3                                          | 78%                | +72%                                   |
| <b>B</b>            | 6           | 780              | 884                    | 1,016                             | 1,059              | 83                     | 12.6                                          | 80%                | +72%                                   |
| <b>C</b>            | 7           | 820              | 886                    | 968                               | 991                | 53                     | 3.0                                           | 35%                | +59%                                   |
| <b>D</b>            | 8           | 860              | 919                    | 987                               | 1,008              | 47                     | 0.5                                           | 15%                | +42%                                   |
| <b>E</b>            | 9           | 900              | 958                    | 1,025                             | 1,048              | 46                     | 0.0                                           | 0%                 | +31%                                   |

| Risk selector $\theta$ | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 |
|------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Recommended option     | A   | A   | A   | C   | C   | C   | C   | C   | C   |

### Reading check

**R1. Which option has the lowest "1-in-20 day" cost?**

☐ N ☐ A ☐ B ☒ C ☐ D ☐ E

**R2. With the selector set to 0.7, which option is recommended?**

☐ A ☐ B ☒ C ☐ D ☐ E

**R3. How many technicians does the option with the lowest average cost use?**

☐ 5 ☒ 6 ☐ 7 ☐ 8

**D4. Which column weighed most in your decision? (one only)**

☐ Technicians ☐ Typical-day cost  
☐ Average cost (40 days) ☐ Average cost of the 12 worst days  
☒ "1-in-20 day" cost ☐ Day-to-day variability  
☐ Overtime of the most loaded technician (t.u.) ☐ Days with overtime  
☐ Cost increase under extreme conditions ☐ selector row

### Your decision

**D1. Which option would you adopt for tomorrow? (one only)**

☐ N ☐ A ☐ B ☒ C ☐ D ☐ E

**D2. If you used the selector, which value would you set?**

☐ 0.1 ☐ 0.2 ☐ 0.3 ☐ 0.4 ☐ 0.5 ☒ 0.6 ☐ 0.7 ☐ 0.8 ☐ 0.9 ☐ I would not use it

**D3. Main reasons for your choice (at most 2)**

☐ lowest typical-day cost ☐ lowest average cost ☒ cheaper worst days  
☐ lower day-to-day variability ☐ fewer technicians ☒ less overtime for the most loaded technician  
☐ better under extreme conditions ☐ I followed the selector ☐ other

**D5. How confident are you in your decision?**

not at all ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒ 6 ☐ 7 completely

## Case 5 — Tomorrow's schedule (several options and selector)

40 work orders must be assigned to technicians for tomorrow. Regular shift: 100 time units (t.u.) per technician. Each activated technician costs 40 cost units (c.u.); each t.u. of work costs 1 c.u.; each t.u. of overtime costs an extra 1.5 c.u. Work-order durations are uncertain and vary by zone: every option was simulated on 40 plausible days plus 12 extreme days.

| Option              | Technicians | Typical-day cost | Average cost (40 days) | Average cost of the 12 worst days | "1-in-20 day" cost | Day-to-day variability | Overtime of the most loaded technician (t.u.) | Days with overtime | Cost increase under extreme conditions |
|---------------------|-------------|------------------|------------------------|-----------------------------------|--------------------|------------------------|-----------------------------------------------|--------------------|----------------------------------------|
| <b>N (standard)</b> | 6           | 796              | 951                    | 1,127                             | 1,196              | 121                    | 20.4                                          | 85%                | +79%                                   |
| <b>P</b>            | 11          | 996              | 1,068                  | 1,152                             | 1,182              | 59                     | 0.0                                           | 0%                 | +31%                                   |
| <b>Q</b>            | 10          | 956              | 1,028                  | 1,112                             | 1,142              | 59                     | 0.0                                           | 0%                 | +37%                                   |
| <b>R</b>            | 7           | 836              | 928                    | 1,049                             | 1,117              | 79                     | 5.9                                           | 55%                | +72%                                   |
| <b>S</b>            | 9           | 916              | 988                    | 1,072                             | 1,105              | 59                     | 0.0                                           | 2%                 | +40%                                   |
| <b>T</b>            | 8           | 876              | 951                    | 1,039                             | 1,088              | 61                     | 1.1                                           | 15%                | +56%                                   |

| Risk selector $\theta$ | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 |
|------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Recommended option     | R   | R   | R   | R   | R   | R   | T   | T   | T   |

### Reading check

**R1. Which option has the lowest "1-in-20 day" cost?**

☐ N ☐ P ☐ Q ☐ R ☐ S ☒ T

**R2. With the selector set to 0.7, which option is recommended?**

☐ P ☐ Q ☐ R ☐ S ☒ T

**R3. How many technicians does the option with the lowest average cost use?**

☐ 6 ☒ 7 ☐ 8 ☐ 9

**D4. Which column weighed most in your decision? (one only)**

☐ Technicians ☐ Typical-day cost  
☐ Average cost (40 days) ☐ Average cost of the 12 worst days  
☒ "1-in-20 day" cost ☐ Day-to-day variability  
☐ Overtime of the most loaded technician (t.u.) ☐ Days with overtime  
☐ Cost increase under extreme conditions ☐ selector row

### Your decision

**D1. Which option would you adopt for tomorrow? (one only)**

☐ N ☐ P ☐ Q ☐ R ☐ S ☒ T

**D2. If you used the selector, which value would you set?**

☐ 0.1 ☐ 0.2 ☐ 0.3 ☐ 0.4 ☐ 0.5 ☐ 0.6 ☒ 0.7 ☐ 0.8 ☐ 0.9 ☐ I would not use it

**D3. Main reasons for your choice (at most 2)**

☐ lowest typical-day cost ☐ lowest average cost ☒ cheaper worst days  
☐ lower day-to-day variability ☐ fewer technicians ☐ less overtime for the most loaded technician  
☐ better under extreme conditions ☒ I followed the selector ☐ other

**D5. How confident are you in your decision?**

not at all ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5 ☐ 6 ☐ 7 completely

## Part 5.1 — Two ways of presenting the options (table and selector only)

40 work orders must be assigned to technicians for tomorrow. Regular shift: 100 time units (t.u.) per technician. Each activated technician costs 40 cost units (c.u.); each t.u. of work costs 1 c.u.; each t.u. of overtime costs an extra 1.5 c.u. Work-order durations are uncertain and vary by zone: every option was simulated on 40 plausible days plus 12 extreme days.

| Option              | Technicians | Typical-day cost | Average cost (40 days) | Average cost of the 12 worst days | “1-in-20 day” cost | Day-to-day variability | Overtime of the most loaded technician (t.u.) | Days with overtime | Cost increase under extreme conditions |
|---------------------|-------------|------------------|------------------------|-----------------------------------|--------------------|------------------------|-----------------------------------------------|--------------------|----------------------------------------|
| <b>N (standard)</b> | 5           | 751              | 882                    | 1,070                             | 1,131              | 123                    | 23.9                                          | 95%                | +73%                                   |
| <b>A</b>            | 6           | 761              | 837                    | 968                               | 1,024              | 83                     | 7.9                                           | 65%                | +69%                                   |
| <b>B</b>            | 7           | 801              | 853                    | 940                               | 966                | 56                     | 1.8                                           | 30%                | +53%                                   |
| <b>C</b>            | 8           | 841              | 889                    | 968                               | 992                | 52                     | 0.0                                           | 0%                 | +37%                                   |
| <b>D</b>            | 9           | 881              | 929                    | 1,008                             | 1,032              | 52                     | 0.0                                           | 0%                 | +31%                                   |

| Risk selector $\theta$ | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 |
|------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Recommended option     | A   | A   | A   | A   | A   | B   | B   | B   | B   |

**V1.1. With the selector at 0.8, how much higher is the typical-day cost than N?**

☐ less than 20 c.u.    ☒ 20–60 c.u.    ☐ 60–100 c.u.    ☐ more than 100 c.u.

**V1.2. With the selector at 0.8, how much lower is the “1-in-20 day” cost than N?**

☐ less than 50 c.u.    ☐ 50–100 c.u.    ☒ 100–200 c.u.    ☐ more than 200 c.u.

**V1.3. How easy was it to answer with this presentation?**

very difficult    ☐ 1    ☐ 2    ☐ 3    ☒ 4    ☐ 5    very easy

## Part 5.2 — Two ways of presenting the options (table, selector and summary)

40 work orders must be assigned to technicians for tomorrow. Regular shift: 100 time units (t.u.) per technician. Each activated technician costs 40 cost units (c.u.); each t.u. of work costs 1 c.u.; each t.u. of overtime costs an extra 1.5 c.u. Work-order durations are uncertain and vary by zone: every option was simulated on 40 plausible days plus 12 extreme days.

| Option              | Technicians | Typical-day cost | Average cost (40 days) | Average cost of the 12 worst days | “1-in-20 day” cost | Day-to-day variability | Overtime of the most loaded technician (t.u.) | Days with overtime | Cost increase under extreme conditions |
|---------------------|-------------|------------------|------------------------|-----------------------------------|--------------------|------------------------|-----------------------------------------------|--------------------|----------------------------------------|
| <b>N (standard)</b> | 6           | 767              | 888                    | 1,035                             | 1,063              | 106                    | 15.9                                          | 88%                | +63%                                   |
| <b>A</b>            | 6           | 767              | 850                    | 965                               | 988                | 81                     | 9.0                                           | 62%                | +64%                                   |
| <b>B</b>            | 7           | 807              | 861                    | 936                               | 941                | 54                     | 0.8                                           | 20%                | +49%                                   |
| <b>C</b>            | 8           | 847              | 898                    | 969                               | 979                | 52                     | 0.0                                           | 2%                 | +33%                                   |
| <b>D</b>            | 9           | 887              | 938                    | 1,008                             | 1,019              | 52                     | 0.0                                           | 0%                 | +27%                                   |

| Risk selector $\theta$ | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 |
|------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Recommended option     | A   | A   | A   | A   | B   | B   | B   | B   | B   |

Automatic summary. With the selector at 0.8 the system recommends option B: 7 technicians instead of the 6 of the standard recommendation N. On a typical day it costs 40 c.u. more (5%); on a “1-in-20 day” it costs 122 c.u. less (12%). The average overtime of the most loaded technician falls from 15.9 to 0.8 t.u. and the share of days with overtime from 88% to 20%. Day-to-day variability falls from 106 to 54: days become more predictable, which does not mean equal workloads across technicians.

### V2.1. With the selector at 0.8, how much higher is the typical-day cost than N?

☐ less than 20 c.u.    ☒ 20–60 c.u.    ☐ 60–100 c.u.    ☐ more than 100 c.u.

### V2.2. With the selector at 0.8, how much lower is the “1-in-20 day” cost than N?

☐ less than 50 c.u.    ☐ 50–100 c.u.    ☒ 100–200 c.u.    ☐ more than 200 c.u.

### V2.3. How easy was it to answer with this presentation?

very difficult    ☐ 1    ☐ 2    ☐ 3    ☐ 4    ☒ 5    very easy

## Part 6 — Final questionnaire

Think of the presentation with several options and the risk selector. 1 = strongly disagree, 5 = strongly agree.

|     | Statement                                                                                | 1                                   | 2                                   | 3                        | 4                                   | 5                                   |
|-----|------------------------------------------------------------------------------------------|-------------------------------------|-------------------------------------|--------------------------|-------------------------------------|-------------------------------------|
| S1  | I think that I would like to use this tool frequently.                                   | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| S2  | I found the tool unnecessarily complex.                                                  | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            |
| S3  | I thought the tool was easy to use.                                                      | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| S4  | I think that I would need the support of a technical person to be able to use this tool. | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            |
| S5  | I found the various functions in this tool were well integrated.                         | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| S6  | I thought there was too much inconsistency in this tool.                                 | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            |
| S7  | I would imagine that most people would learn to use this tool very quickly.              | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| S8  | I found the tool very cumbersome to use.                                                 | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            |
| S9  | I felt very confident using the tool.                                                    | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| S10 | I needed to learn a lot of things before I could get going with this tool.               | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            |
| G1  | I understood what the risk selector does.                                                | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| G2  | The table contains the information I actually use to decide.                             | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| G3  | Having several options helped me more than it confused me.                               | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| G4  | To check attention, please click box 4 in this row.                                      | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| G5  | I would trust these numbers if they were computed on my company's data.                  | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| G6  | I would rather receive a single recommendation.                                          | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            |
| G7  | The written summary (Part 5) helped me understand the differences between the options.   | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |

### F1. How much extra time would you spend on a daily decision to have several options?

☐ none   ☐ up to 1 minute   ☒ 1–3 minutes   ☐ 3–5 minutes   ☐ more than 5 minutes

### F2. What else would you like to see? (several answers allowed)

☒ workload of each technician   ☐ map of the zones   ☐ comparison with real past days  
☒ automatic written explanation   ☐ nothing   ☐ other

### F3. (Optional) A one-line comment: .....

End time: ...

*Thank you. Save the file and upload it to the COMPLETED folder with the same file name.*